Sustainability Datathon

Electric Dragon
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Spain stands at a pivotal moment in its energy transition. The European Union's Alternative Fuels Infrastructure Regulation (AFIR) [1] mandates at least one 150 kW fast-charging station every 60 km on the TEN-T core network from 2025, with full coverage by 2030. Spain's own National Energy and Climate Plan (PNIEC) [2] targets 5.5 million electric vehicles on its roads by 2030. Yet as of Early 2026, only ~550,000 EVs are in circulation [C], primarily driven by the bottleneck in long-distance travel and the phenomenon known as "range anxiety."

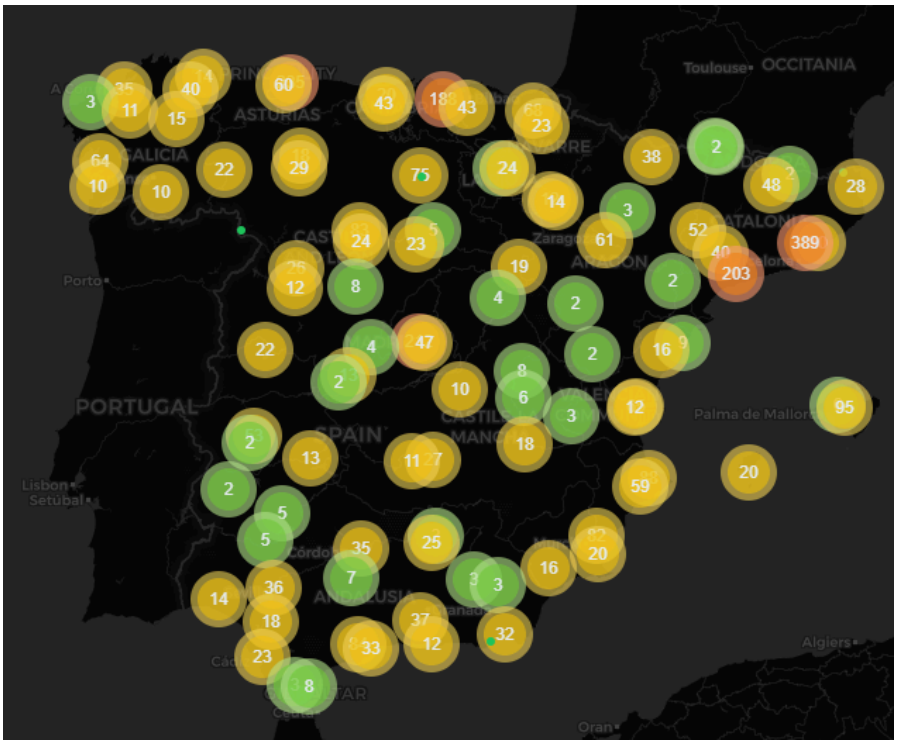
To deliver Iberdrola's Value Proposition: Reliable, fast, green charging on every inter-urban corridor. The solution lies in (executive summary for each)

1. Objective 1: Charging Network Optimization
    a. Maximizing Utilization Rates
    b. Maximize Convenience (Distance between Stations)
    c. Minimizing Capex
    d. Maximizing Profitability
2. Objective 2: Grid Viability Analysis
    a. Grid Constraint - Friction Points
3. Objective 3: Strategy and Value Proposition
    a. Improving the Optimization Model
    b. Addressing the Friction Points
    c. Timeline

Methodology and Analytics

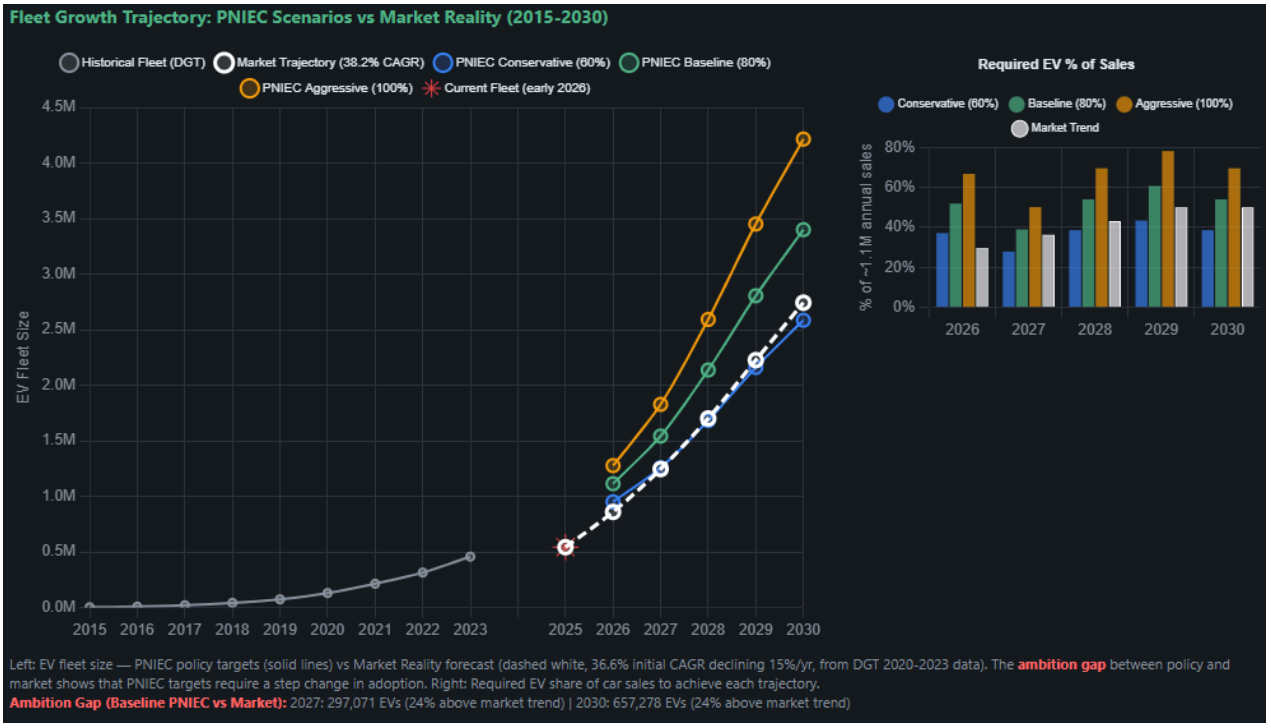
Existing Stations

Data: A + B



Iberdrola -
Partners -

2027 Projected EV Fleet / Traffic Volume



- Forecasting the registrations to identify the distribution of cars per lane.
- Forecast day of week/day per lane (assuming sedan/family car type given 80% volume)
- Customer Profile
    - ____
    - ____
- P80 Assumption

Proposed Charging Stations

1. Minimum - **2**
2. Viability (Business Model)
    - **Revenue**
        - **Pay-per-kWh retail (€0.45-0.65/kWh)**
        - Fleet subscription contracts
        - OCPI roaming fees
        - **EU AFIR/CEF grants (up to 30% CAPEX)**
        - Grid ancillary services (FCR/mFRR)
        - Media & co-branding screens
    - **CAPEX**: Avg. €2.08M total per site
        - Charger hardware €25-40k/unit
        - Grid connection €80-300k/site
        - Civil works €40-120k/site
    - **OPEX**:
        - **Energy procurement (€0.18/kWh avg.)**

        - O&M 5-8% of CAPEX/yr
        - **Land lease €10-30k/site/yr**
        - CPMS platform & NOC
        - Insurance & regulatory
3. Car Type Usage Rate
    - **Vehicle Performance:** Average EV consumption is estimated at **0.155-0.20 kWh/km**, accounting for Spain's diverse topography and real-world driving conditions.
    - EV Autonomy - Average 320–640 km
    - Max-Min Battery % - **80-20%**
4. Regulation - Compliance
    - EU AFIR regulation mandate requiring charging infrastructure every 60 km on the **TEN-T core network** by end-2025
1. Limit Infrastructure - Limit CAPEX, Increase Synergy
    - Iberdrola - Existing Stations
    - B2B Partner - Existing Stations
    - B2B Partner

Proposed Charging Points

1. Chargers Points per Site (n)
    - Standard power capacity per charger is fixed at **150 kW**.
    - Design by Min, Avg, Max - 3rd Quartile - Time Trend - Daily level of Car Demand per Route. 80% Volume

        - **Service Levels:** Models assume an EU-aligned service ratio of **7 to 10 EVs per public charging point**.

    - **Constraints**
        - **Sqm Capacity per site** - parking lot
        - **Power Availability and Capacity per site** - from the grid
        - **Viability per site** - viable.

    - Grid Status (TBD)

5. **Friction Points** - Moderate and Congested
    - Locations
    - Distribution
    - Estimated Demand

6. **Global Network KPIs:**
    - How many stations are proposed (2)
    - How many existing stations are there? (Dataset)
    - How many friction points (3)
    - 2027 Projected EV fleet (Datos.gob.es) Traffic Volume
        - Forecasting the registrations to identify the distribution of cars per lane.

        - Forecast day of week/month per lane.

Strategic Proposals

- **Increase Demand**
    - Adoption
        - Target Market: Buying electric cars (25-35 yrs old, young professionals)
    - Every 50 kms
- **Increase Capacity / Supply**
    - **Regulatory Efficiency:** It is assumed that the **"Plan Auto+ 2026"** and associated reforms will reduce permitting and activation cycles to a 30–60 day window.
    - Solar Charging
        - **Energy Mix:** The network assumes a **100% green energy supply** backed by Guarantees of Origin (GdO), consistent with Iberdrola's vertical integration strategy.
    - Target Market: B2B2C:
        - Partners
        - Residentials?
    - B2C is only in stable demand.
    - B2B2C Temporary for Demand Spikes
- **Corridor Reinforcement:** Prioritize deployment along the **Mediterranean Corridor (AP-7)** and the primary **Radial Routes (A-1 to A-6)** to secure seamless transit for both tourists and heavy logistics.
- **Phased Rollout:** Implement a transition from "land grabbing" for new sites to **"activating" the current 16,340 non-operational points** through proactive engagement with grid distributors.
- **Grid Flexibility via V2G:** Incentivize the installation of **bidirectional (V2G) hardware** to allow EV fleets to act as "batteries on wheels," potentially mitigating 15-25% of peak load increases on fragile grids.
- **Corporate Value Proposition:** Introduce **Charging-as-a-Service (CaaS)** models and unified fleet billing to reduce CAPEX for businesses electrifying their delivery operations.
- **Digital Transparency:** Advocate for a **national digital permitting portal** to standardize procedures across autonomous communities and reduce the "zombie" charger rate.

- **Energy Efficiency Consulting:** Using AI to optimize charging schedules based on real-time electricity prices and building load constraints.

5. Timeline

Continue building, every day and together, a more electric, healthy and accessible energy model.

Company Values.
We are Sustainable Energy
We are Driving Force
We are Integrating Force

Review of Related Literature

MAIN REFERENCES PROVIDED:

A. Road Routes | Ministry of Transport and Sustainable Mobility
B. Electric vehicle charging points - Datasets | National Access Point for Traffic and Mobility
C. Route to electrification: Deciphering the growth of electric vehicles in Spain through data analytics | datos.gob.es
D. IBERDROLA: i-DE (Consumption capacity map)
E. ENDESA: e-distribución (Network capacity nodes)
F. VIESGO: Interactive grid map
G. Vehicle registrations in Spain (DGT)
H. Electric car charging points | datos.gob.es

ADDITIONAL DATA REFERENCES

I. 2025 Iberdrola Annual Financial Report
J.

ADDITIONAL DATA REFERENCES

EU REGULATORY FRAMEWORK — AFIR & TEN-T

[1] European Parliament and Council of the European Union (2023). *Regulation (EU) 2023/1804 on the deployment of alternative fuels infrastructure (AFIR), repealing Directive 2014/94/EU.* Official Journal of the European Union, L 234, 22 September 2023. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32023R1804

[2] Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO), Government of Spain (2023). *Draft Updated Plan Nacional Integrado de Energía y Clima (PNIEC) 2023–2030.* European Commission NECP Submission. https://commission.europa.eu/system/files/2023-07/EN_SPAIN%20DRAFT%20UPDATED%20NECP.pdf

[3] European Alternative Fuels Observatory (EAFO) (2025). *Spain: BEV passenger car sales increased by 7.8% in 2024 — PNIEC target update.* EAFO Country News. https://alternative-fuels-observatory.ec.europa.eu/general-information/news/spain-bev-passenger-car-sales-increased-78-2024

[4] Iberdrola España (2025). *Electric mobility in Spain — 600,000 EVs and 10,000 charging points.* Iberdrola España Corporate Website. https://www.iberdrolaespana.com/about-us/business-lines/energy-solutions-customers-spain/electric-mobility-spain

[5] International Energy Agency (IEA) (2024). *Regulation (EU) 2023/1804 Alternative Fuels Infrastructure Regulation — Policy Overview.* IEA Policies Database. https://www.iea.org/policies/27375-regulation-eu-20231804-alternative-fuels-infrastructure-regulation

TECHNICAL STANDARDS — OCPP, OCPI & ISO 15118

[6] Open Charge Alliance (OCA) (2025). *Open Charge Point Protocol (OCPP) — version history and OCPP 2.0.1 / OCPI 2.2.1 specifications.* Open Charge Alliance. https://openchargealliance.org/protocols/open-charge-point-protocol/

EU FUNDING — CEF TRANSPORT / AFIF

[7] European Climate, Infrastructure and Environment Executive Agency (CINEA) (2025). *Alternative fuels: €422 million of EU funding to boost zero-emission mobility — AFIF first cut-off results.* CINEA News, 6 February 2025. https://cinea.ec.europa.eu/news-events/news/alternative-fuels-eu422-million-eu-funding-boost-zero-emission-mobility-2025-02-06_en

[8] European Alternative Fuels Observatory (EAFO) (2026). *Spain's 2026 e-Mobility Incentives: Auto+ and Plan Auto 2030 (MOVES III successor).* EAFO Country News. https://alternative-fuels-observatory.ec.europa.eu/general-information/news/spains-2026-e-mobility-incentives-auto

BUSINESS MODEL METHODOLOGY

[9] Osterwalder, A. & Pigneur, Y. (2010). *Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers.* John Wiley & Sons, Hoboken NJ, ISBN 978-0-470-87641-1.

GRID INFRASTRUCTURE & ENERGY MARKETS

[10] Red Eléctrica de España (REE / ENTSO-E) (2024). *Spanish electricity system: grid connection regulations, peajes de acceso and capacity data.* REE Corporate Website. https://www.ree.es/en

ADDITIONAL SPANISH EV FLEET & MARKET DATA

[11] Anari Energy (2025). *Accelerating the Charge: Strategic landscape of Spain's EV Charging Infrastructure.* Anari Energy Research. https://www.anariev.com/strategic-landscape-of-spain-ev-charging-infrastructure/

CPO FINANCIAL BENCHMARKS & COMPETITOR DATA

[12] AInvest (2025). *Iberdrola's Strategic Expansion in Electric Vehicle Charging Infrastructure: A Catalyst for Sustainable Mobility and Investment Growth.* AInvest Research, August 2025. https://www.ainvest.com/news/iberdrola-strategic-expansion-electric-vehicle-charging-infrastructure-catalyst-sustainable-mobility-investment-growth-2508/

ENERGY PRICING — OMIE

[13] Operador del Mercado Ibérico de Energía (OMIE) (2025). *Iberian electricity market — wholesale price data and forward curves.* OMIE Market Data Portal. https://www.omie.es/en

ADDITIONAL ROAD TRAFFIC & CORRIDOR DATA

[14] Dirección General de Tráfico (DGT), Ministerio del Interior, Government of Spain (2024). *Anuario estadístico de accidentes 2023 — Mapa de tráfico: intensidades medias diarias en la red de carreteras del Estado.* DGT Open Data. https://www.dgt.es/menusecundario/dgt-en-cifras/dgt-en-cifras-resultados/dgt-en-cifras-detalle/?id=00216

ADDITIONAL EV CHARGING MARKET ANALYSIS

[15] Roland Berger (2025). *EV Charging Index 2025: Expert insights from China — global charging market benchmarks and utilisation data.* Roland Berger Publications. https://www.rolandberger.com/en/Insights/Publications/EV-Charging-Index-2025-Expert-insights-from-China.html

[16] Nicholas, M. & Wappelhorst, S. (ICCT) (2021). *Spain's electric vehicle charging infrastructure requirements through 2030.* ICCT Working Paper 2021-03, January 2021. https://theicct.org/sites/default/files/publications/Spain-EV-charging%20infra-jan2021.pdf

Exhibit 1: Business Model Canvas

The following canvas maps Iberdrola's value creation, adapted to the EV fast-charging context on Spain's inter-urban network.

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
DGT / MITECO [1] Automakers (Stellantis, VW, Hyundai) [3] AP motorway operators REE (grid capacity) [10] EU CEF Transport fund [7] Fleet operators (Cabify, Correos) [11]	Site selection & grid feasibility CPMS platform ops (OCPP 2.0.1) [6] Demand forecasting & load mgmt Roaming interoperability (OCPI 2.2.1) [6] 24/7 NOC monitoring AFIR regulatory compliance [1]	**Reliable, fast, green charging on every inter-urban corridor** ≥150 kW DC per connector [1] ≥99% uptime SLA 100% renewable energy (GO certs) [4] Plug-and-charge (ISO 15118) [6] Multi-standard: CCS2, CHAdeMO, AC V2G readiness for grid services	Iberdrola app (loyalty & history) [4] B2B fleet subscription contracts Plug-and-charge seamless UX [6] 24/7 helpline & remote unlock OEM white-label portals [3] OCPI roaming (zero-friction) [6]	**Inter-urban BEV travellers** EV fleet & logistics operators [11] Ride-hail & taxi Tourism & rental vehicles [2] OEM / mobility partners (B2B) [3] Grid operators (V2G/DSR, future)
Key Resources	**Channels**	**Cost Structure**	**Revenue Streams**	
Grid & HV infrastructure (AT/MT) [10] In-house CPMS software platform [4] Renewable generation (PV self-supply) [4] 150+ owned/leased site portfolio Engineering & operations workforce EV charging brand & regulatory licences	Iberdrola EV app (iOS/Android) [4] OCPI roaming (Zunder, Ionity) [6] OEM in-car navigation integration [3] Google Maps / PlugShare Fleet portals & API billing [11] Motorway DGT co-branded signage [2]	CAPEX: charger hardware €25–40k/unit | Grid connection €80–300k/site [12] | Civil works €40–120k/site | Avg. €2.08M total per site OPEX: Energy procurement (€0.18/kWh avg.) [13] | O&M 5–8% of CAPEX/yr | Land lease €10–30k/site/yr | CPMS platform & NOC | Insurance & regulatory	**Pay-per-kWh retail (€0.45–0.65/kWh) [13]** Fleet subscription contracts [11] OCPI roaming fees [6] EU AFIR/CEF grants (up to 30% CAPEX) [7,8] Grid ancillary services (FCR/mFRR) [10] Media & co-branding screens	

NAP Stations by Number of Connectors

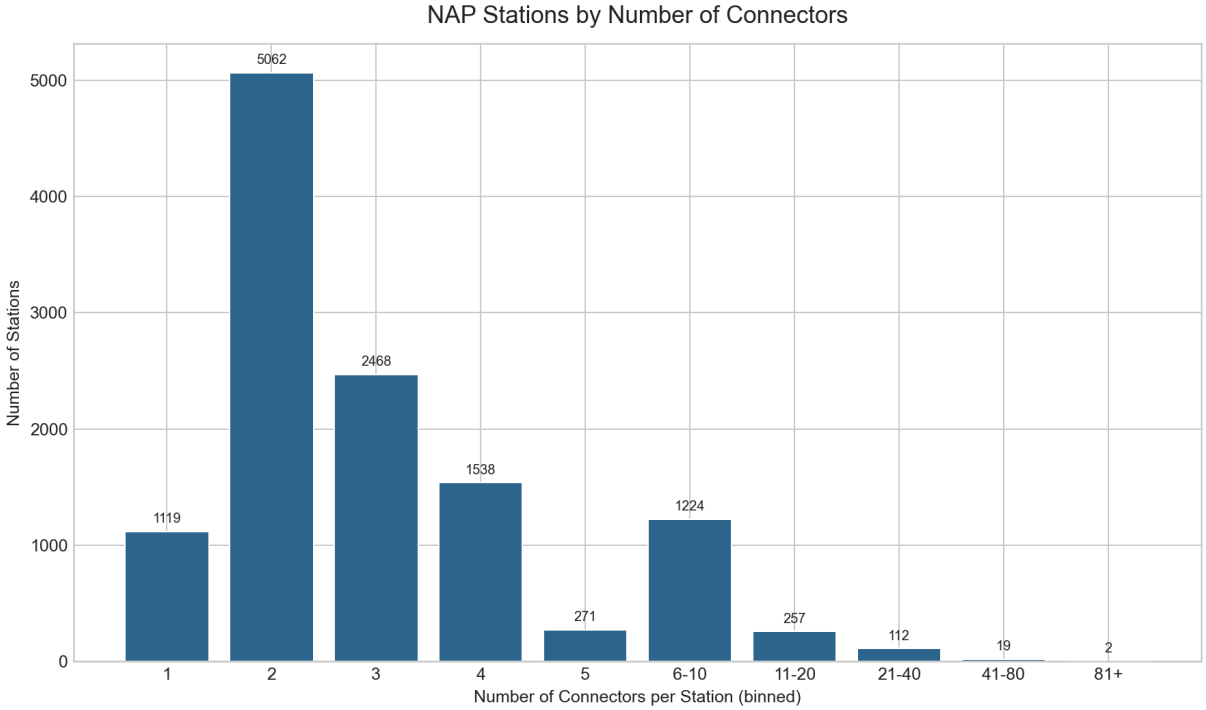


Exhibit 2: BI Visualization

Descriptive
Diagnostic
Predictive
Prescriptive

1. Map
2. City/Route List - with Factor Rating Method
3. Flexible Filters - for sensitivity

Based on the provided sources, the Spanish Electric Vehicle (EV) charging infrastructure industry is undergoing a transformative scale-up, reaching over **53,000 public points by the end of 2025**. Below is a Porter's Five Forces and Value Chain analysis, followed by the critical variables for each entity.

Porter's Five Forces Analysis

Force	Intensity	Key Factors
Competitive Rivalry	High	Intense competition between traditional utilities (**Iberdrola, Endesa**), oil companies (**Repsol, Cepsa**), and independent operators (**Zunder, Wenea, Tesla**). There is a race for "land grabbing" in strategic corridors.
Threat of New Entrants	Moderate	**High CAPEX** (€45k–€100k per fast charger) and bureaucratic delays (6–24 month permitting) act as barriers. However, **EU funding** and government subsidies lower the risk for well-capitalized tech firms.
Bargaining Power of Suppliers	High	Infrastructure depends on a few **Grid Operators (DSOs)** for connection. Equipment vendors (ABB, Siemens, Wallbox) are vital for **OCPP-compliant hardware**. Wholesale energy price volatility impacts operator margins.
Bargaining Power of Buyers	Moderate	Consumers currently face fragmented apps and payment platforms. As the network matures, **AFIR mandates** for price transparency and interoperability increase buyer power.
Threat of Substitutes	Low	While ICE vehicles are the current alternative, the **2035 ban** on new combustion sales makes electrification inevitable. Hydrogen is a niche substitute for heavy-duty transport.

Value Chain Analysis

The industry value chain translates energy into mobility through several primary and support activities:

1. Inbound Logistics (Equipment & Energy Sourcing):
    - **Hardware Manufacturing:** Production of AC/DC chargers, power modules, and smart meters.
    - **Energy Generation:** Procuring renewable energy via **PPAs** and ensuring **Guarantees of Origin (GdO)** for green certification.
2. Operations (Infrastructure Deployment):
    - **Site Acquisition:** Identifying high-traffic interurban routes and highway corridors.
    - **Permitting & Installation:** Navigating municipal approvals and physical deployment.

- **Grid Connection:** Securing access to power nodes from distributors.

3. Outbound Logistics (Charging Services):
   - **Charging Point Operation (CPO):** Technical hardware management and maintenance.
   - **e-Mobility Service Provider (eMSP):** Providing the user interface, billing, and roaming management.
4. Marketing & Sales:
   - **Loyalty Programs:** Integrated apps (e.g., Repsol's Waylet) and corporate "Charging-as-a-Service" (CaaS) models.
5. Service:
   - **Network Reliability:** Managing uptime and providing customer support for charging sessions.

Critical Variables for Value Chain Entities

For a robust analysis of the entities within this chain, the following variables are essential:

- Manufacturers:
  - **Compliance Standards:** Adoption of OCPP 2.0.1 and ISO 15118 for V2G readiness.
  - **Scalability:** Ability to produce modular power stacks for ultra-fast (>250 kW) charging.
- Grid Operators (DSOs):
  - **Available Capacity:** Real-time MW capacity at substation levels.
  - **Connection Lead Times:** The time required to activate installed hardware (currently a major bottleneck).
- Charging Point Operators (CPOs):
  - **Utilization Rate:** Sessions per day; critical for ROI in low-traffic vs. urban zones.
  - **Uptime/Reliability:** Percentage of operational units (currently below 92% in some areas).
  - **Installation CAPEX:** Total cost per site, including grid fees and hardware.
- Energy Retailers:
  - **Retail/Wholesale Spread:** Managing the margin between volatile market prices and consumer tariffs.
  - **Demand Response Capacity:** MW potential for grid balancing via **V2G**.

- End-Users (Consumers & Fleets):
    - **EV Adoption Rate:** Regional penetration index (e.g., Madrid at 15.6% vs. national average).
    - **Charging Speed:** Demand for <30 minute "refueling" parity with diesel.

Would you like me to create a **tailored report** summarizing this analysis with strategic recommendations for a specific stakeholder, such as a CPO or a utility company? Alternatively, I can generate an **infographic** to visualize the growth trends and variables across the value chain.

Based on the provided sources, Iberdrola's competitive advantage, brand promise, and market positioning are defined by its scale, its vertical integration as a renewable energy producer, and its strategic focus on nationwide connectivity.

Competitive Advantage

Iberdrola maintains several critical advantages that distinguish it from oil majors and independent operators:

- **Network Scale and Leadership:** Iberdrola is the operator with the largest number of public charging points in Spain, surpassing **11,000 operational points** by late 2025. This extensive footprint provides a volume-based advantage that smaller competitors cannot easily match.
- **Vertical Integration and Green Energy:** As a national utility, Iberdrola leverages its own generation capacity, guaranteeing that **100% of the energy supplied** at its points is green, backed by Certificates of Guarantee of Origin (GdO). This allows them to offer a "pure" sustainability value proposition compared to multi-fuel competitors.
- **Strategic Alliances:** The company has built a massive ecosystem through over **50 agreements** with diverse partners, including automotive OEMs (Volvo, SEAT, Volkswagen, Renault, Toyota), commercial chains (IKEA, McDonald's), and public institutions (EMT Madrid).
- **Technological Maturity:** Their public charging app is cited as the **highest-rated in the sector**, offering real-time verification, geolocating, and mobile payment.

Brand Promise

Iberdrola's brand promise centers on enabling the transition to a sustainable future through accessibility and environmental integrity:

- **"Connecting the present with the energy of the future"**: This core slogan reflects their role in bridging the gap between current infrastructure needs and a fully decarbonized transport model.
- **Autonomy and Reliability:** They promise a **"worry-free" journey** across Spain by committing to at least one fast-charging station every **50 km** on main corridors.
- **Renewable Commitment:** A fundamental part of their promise is ensuring that every kilometer driven using their network is **CO2-free**, leveraging their wind and solar assets.

Market Positioning

Iberdrola positions itself as the **dominant, high-quality, and environmentally sustainable choice** in the Spanish EV market:

- **The "Total Coverage" Leader:** Unlike niche players focusing only on ultra-fast highway transit or urban hubs, Iberdrola positions itself as a **universal provider** covering high-traffic corridors, metropolitan zones, and private residential/corporate segments.

- **Sustainability Benchmark:** They position themselves at the center of the "Circular Economy" and "Green Transition," often being the first choice for corporate fleet electrification due to their irrefutable green certifications (GdOs).
- **Reliable Infrastructure Backbone:** By aiming for **200,000 chargers by 2030** (across public and private spheres), they position themselves not just as a service provider, but as the **physical backbone** of the Spanish electrification strategy.
- **Balanced Pricing Strategy:** They maintain a mid-market pricing position (typically **€0.39–€0.45/kWh** for fast charging), positioning themselves as more affordable than premium networks like Ionity while offering more extensive coverage than low-cost independent operators.

User Segmentation & Demand Profiles

To ensure the proposed charging network aligns with real-world usage, EV users are segmented into four key personas based on mobility patterns:

- **Daily Commuters:** require convenient, fast-access charging for frequent short trips: small stations near urban peripheries
- **Long-Distance Travelers:** require high-power, reliable charging along highways : fast-charging hubs every 50–60 km
- **Tourist / Seasonal Drivers:** require accessible and intuitive charging during peak periods: scalable infrastructure in high-tourism corridors
- **Fleet / B2B Users:** require cost-efficient, high-uptime charging: dedicated or contracted charging zones

Each proposed station is designed based on the dominant demand profile of its location. This ensures higher utilization rates, better user experience, and more efficient capital allocation.

Financial Allocation & ROI Optimization Layer

Iberdrola plans to invest **€9 billion in the Iberian Peninsula** between 2025–2028,s estimated to allocate approximately €80M annually to EV charging infrastructure in Spain, as part of its broader €2.25B/year Iberian investment program.

At an average assumption cost of ~€100k per charger, this enables the deployment of approximately 800 new charging points per year, making capital prioritization critical.

This highlights a key financial reality:

Return is driven by utilization, not deployment volume.

Therefore, the challenge is not where chargers can be built, but where capital should be deployed to maximize return and network efficiency.

Profit:

Revenue
Cost

Full

Roi
Payback
Have

Need

Closing

We built a decision system that helps Iberdrola allocate capital efficiently across Spain's EV charging network, balancing demand, grid constraints, and financial return.

Technical hub per charging station
Create a lounge

Nonmarket Strategy

IA³

Core Issue

Problem:

Permitting delays and grid bottlenecks are limiting EV infrastructure deployment

This is the anchor issue, particularly critical given the acceleration of EV adoption driven by European regulation.

The European Union has mandated a transition to zero-emission vehicles by 2035 as part of the European Green Deal, with interim targets requiring a 55% reduction in emissions for new cars by 2030. This creates urgent pressure to scale charging infrastructure rapidly.

2. Prioritization of Key Actors

Actor	Power	Interest	Priority
Government	High	High	Critical
DSOs	High	High	Critical
Iberdrola	High	High	Core
Competitors	Medium	High	Secondary
Users	Low	High	Secondary
NGOs	Medium	Medium	Secondary

Key insight:

The strategy must primarily focus on Government and DSOs, as they control regulation and grid access.

3. Strategic Approach

Adapt strategy (react to constraints) is insufficient

Shape strategy (influence environment) is required

Chosen approach:

A shaping nonmarket strategy to actively reduce regulatory and infrastructure constraints

4. Core Strategic Actions

4.1 Regulatory Strategy (Public Policy Engagement)

Objectives:

Accelerate permitting timelines

Standardize national and regional approval processes

Position EV charging infrastructure as critical infrastructure

Actions:

Build industry coalitions with competitors and stakeholders

Use data-driven evidence to support regulatory proposals

Policy context:

EU regulation already imposes penalties (€95 per gram of CO_2/km) for non-compliance

21 EU countries offer tax incentives and 20 offer purchase subsidies

Spain is aligned with EU electrification goals, creating opportunity for policy support

Expected outcome:

Reduction in inactive infrastructure ("zombie chargers") and faster deployment cycles

4.2 Grid Partnership Strategy

Objectives:

Align infrastructure deployment with grid capacity

Reduce friction between demand and supply

Actions:

Co-invest with DSOs in grid upgrades

Prioritize "grid-ready" corridors

Coordinate charger deployment with substation capacity

Expected outcome:

Improved feasibility and scalability of infrastructure deployment

4.3 Coalition Strategy

Objectives:

Reduce stakeholder conflict

Accelerate implementation

Actions:

Partner with:

Government institutions

DSOs

Municipalities

Mobility and infrastructure partners

Expected outcomes:

Faster approvals

Shared investment burden

Increased political and social legitimacy

5. **Leveraging Information**

The data model should be used as a strategic asset to:

Identify demand hotspots

Highlight grid bottlenecks

Justify infrastructure investment

Role of information:

Transforms technical analysis into a policy and decision-making tool

6. **Deployment of Strategic Assets**

Asset	Strategic Use
Grid infrastructure	Prioritize deployment in owned or controlled areas
Capital	Phase investments to reduce risk and improve returns

Data Influence regulatory decisions and planning priorities

7. **Final Strategic Plan**

Issue

Grid and regulatory bottlenecks are limiting EV infrastructure rollout

Strategy

Shape regulation to accelerate deployment

Align infrastructure with grid capacity through partnerships

Build coalitions across public and private stakeholders

Use data to guide both investment and policy decisions

8. Final Strategic Statement

“We recommend that Iberdrola pursue a shaping nonmarket strategy by engaging regulators to reduce permitting delays, partnering with grid operators to align infrastructure deployment with capacity constraints, and leveraging data-driven insights to influence policy and prioritize investment.”

9. Strategic Context (Why This Matters Now)

EU mandates require full transition to zero-emission vehicles by 2035

Interim targets (2030) create immediate infrastructure demand

Member states, including Spain, provide incentives but face uneven execution

Regulatory and grid constraints are now the primary bottleneck, not demand